

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. CHB-504: Physical Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Brief about any three processes of deactivation of a photochemically activated molecule (internal conversion, intersystem crossing, fluorescence, etc.).
- (b) Give two experimental evidences which validate the Debye-Hückel model of electrolyte solutions.
- (c) Explain how the chemical potential of a reference ion changes as a consequence of ion-ion interactions.
- (d) Define fugacity and describe its physical significance.
- (e) Write the relationship between activity and mole fraction.

Question 2.

- (a) What is an ideal-dilute solution? Explain with the help of Henry's and Raoult's laws. [6]
- (b) State: (i) Gibbs-Duhem equation and (ii) Duhem-Margules equation, and explain their thermodynamic significances. [8]

Question 3.

- (a) State the physical significance of fugacity. Describe the Lewis-Randall rule for gas mixtures. [6]
- (b) Describe osmometry and its applications in the determination of molar masses of macromolecules. [8]

Question 4.

- (a) If I_a is the intensity of the light absorbed, derive an expression for the quantum yield of the photodimerization of anthracene (Φ). [8]
- (b) Discuss the influence of pH and temperature, depicting the graph, on the kinetics of enzyme-catalyzed reactions. [6]

Question 5.

- (a) What do you understand by the thickness or radius of the ionic cloud that surrounds a reference ion? Show the variation of the thickness of the ionic cloud as a function of electrolyte concentration and prove that it has the dimensions of length. [8]
- (b) Under which conditions is the Wien effect observed? Explain, giving examples, the increase in the conductivities of electrolytic solutions with the application of high electric fields. [6]